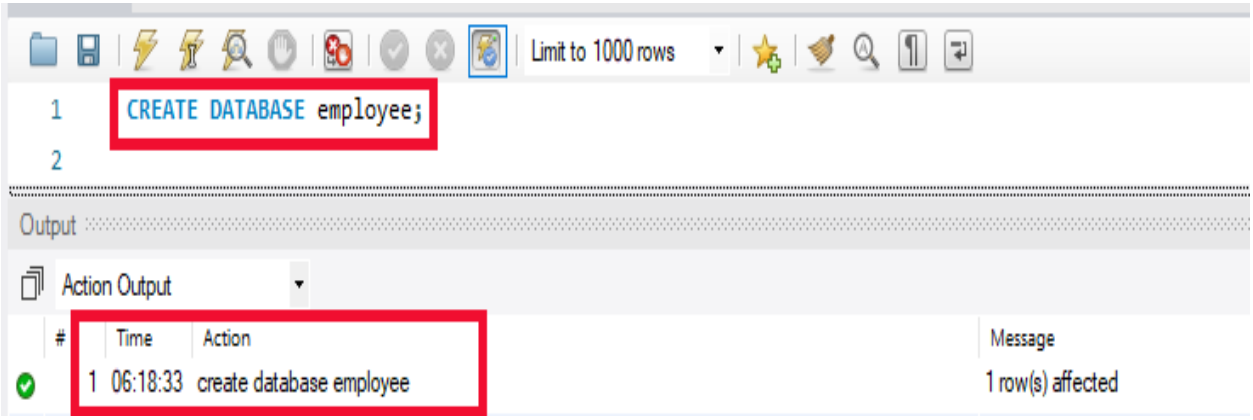
SQL Assignment One

DDL Comments

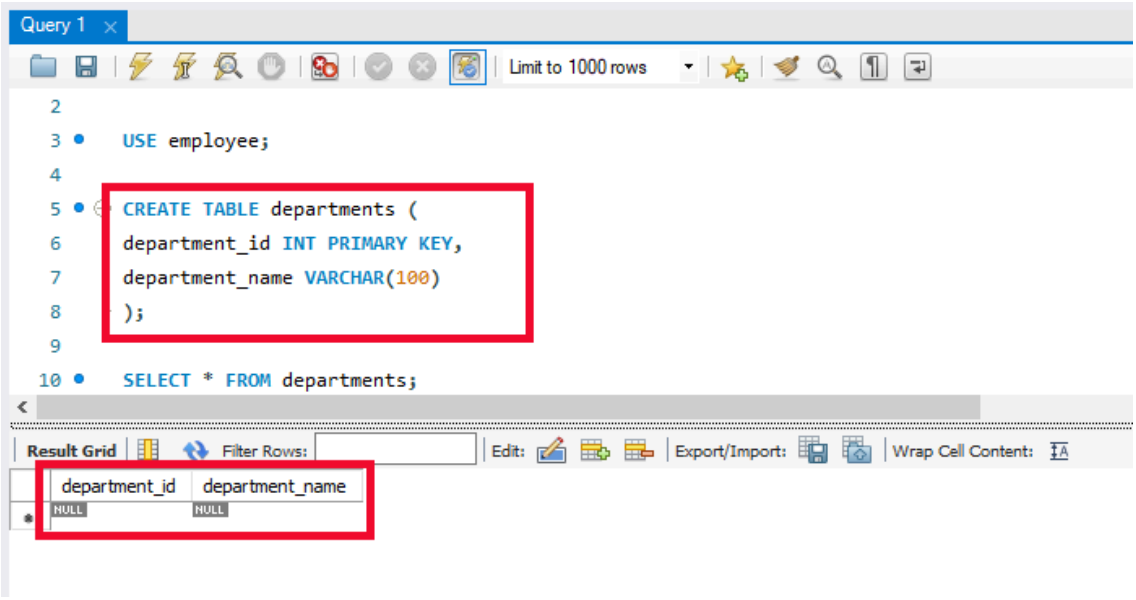
DATABASE & TABLE Creations

Database creation in the name of **employee**

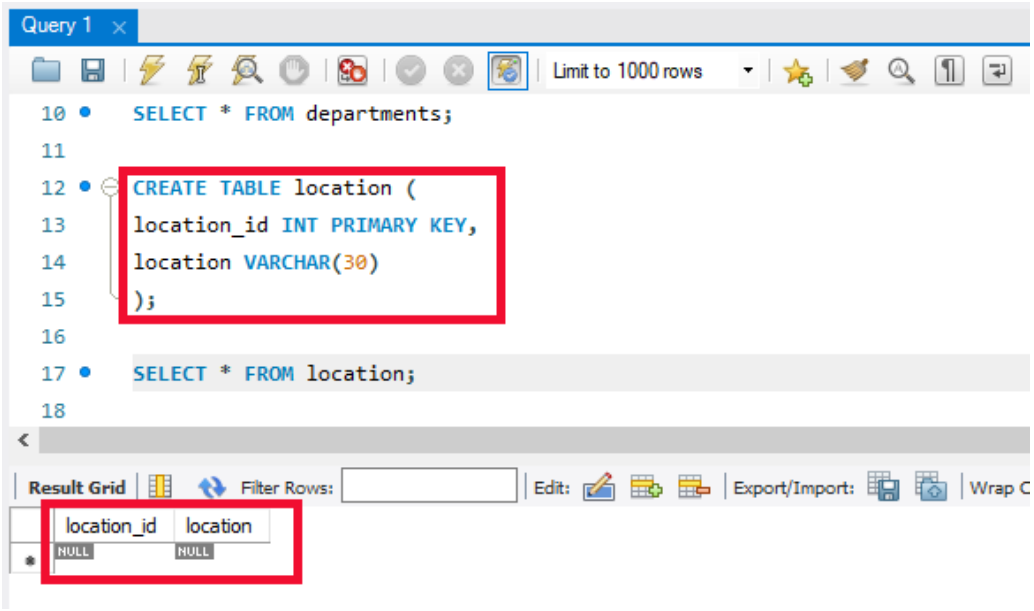


1. Table Creation

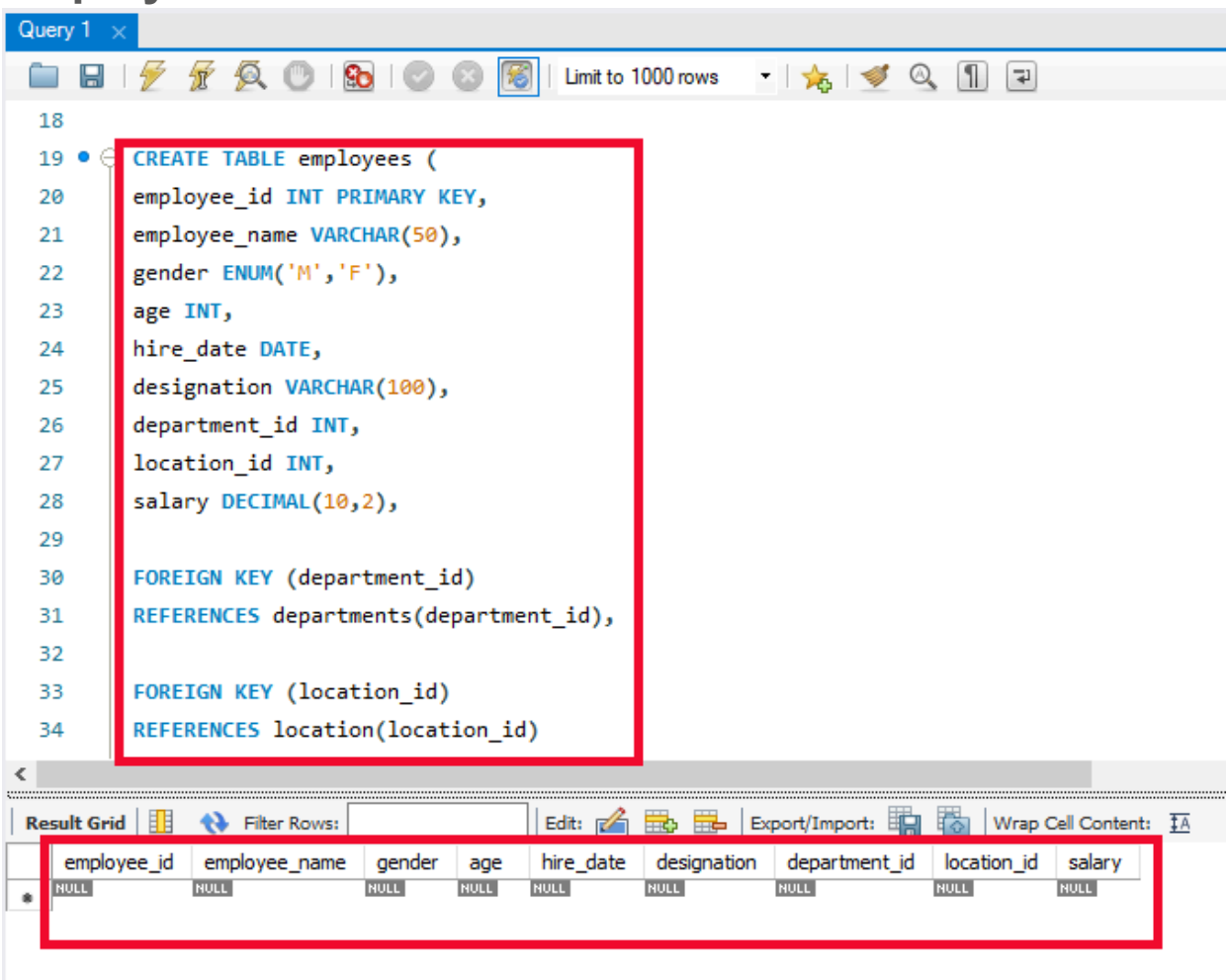
- **Departments** table creation. Here the dept_id was Primary Key. This Primary key will help to connect another table. Like if the employee table was the parent table and the department table was the child table. Both the tables must have the same field dept_id.



❖Location



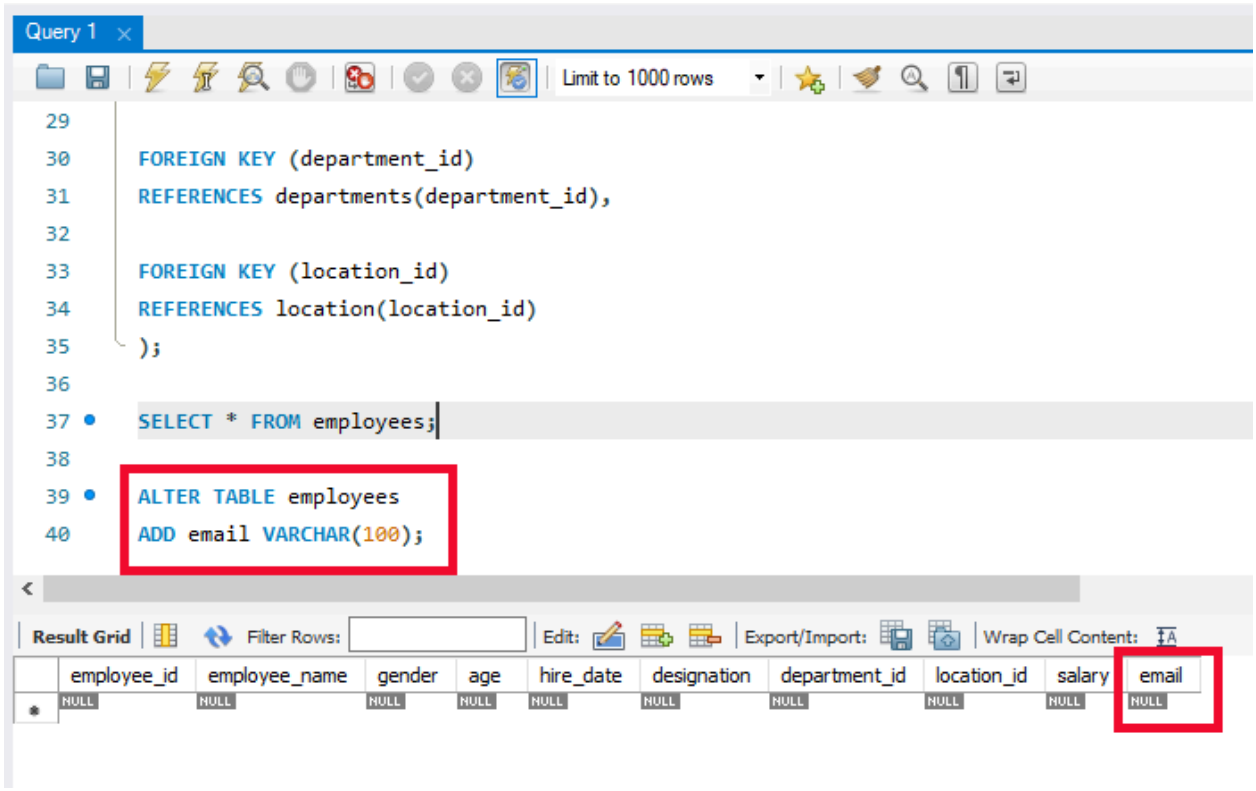
❖Employees



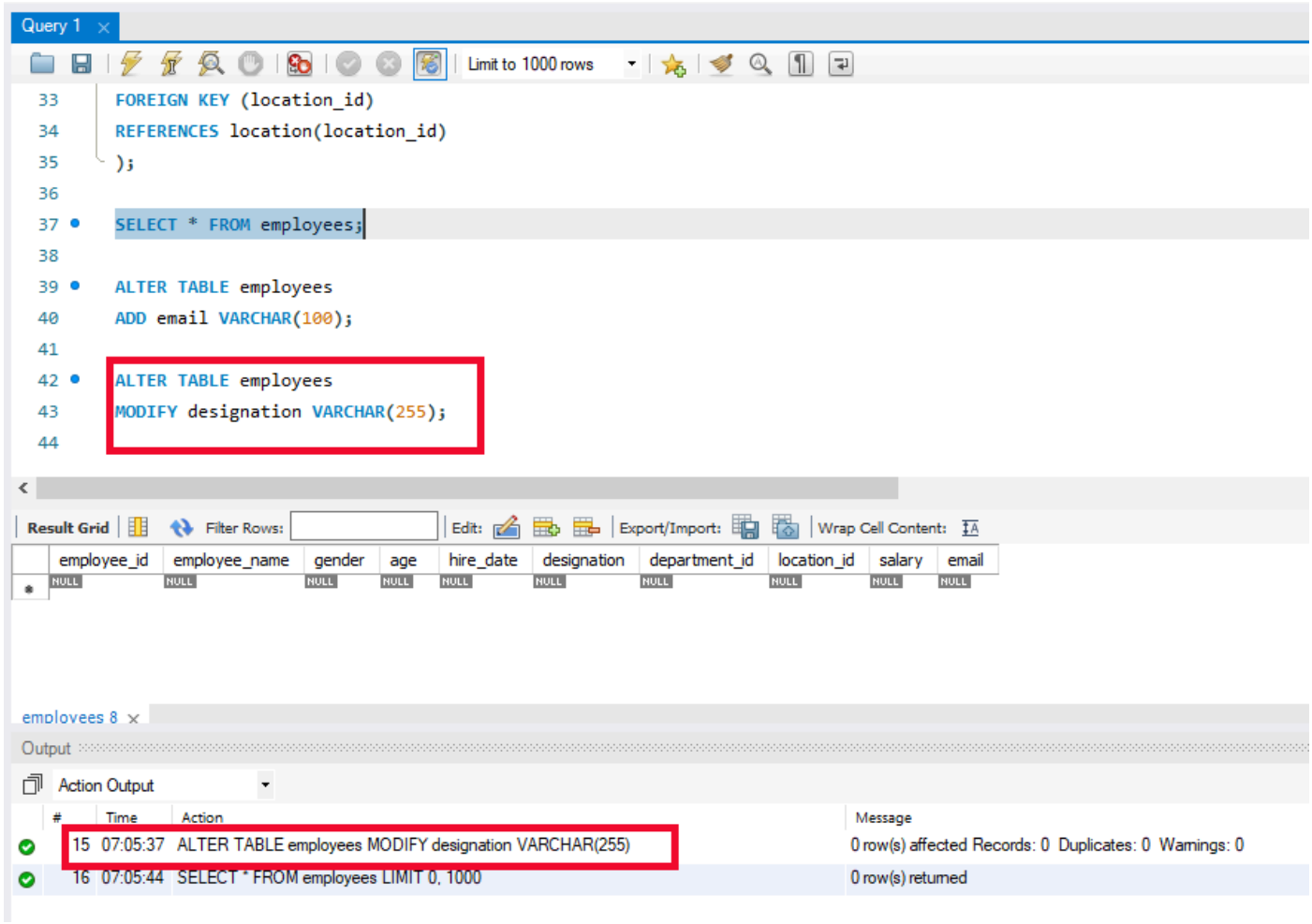
2. Table Alteration

The **ALTER TABLE** statement is used to change the structure of an existing table by adding, modifying, deleting, or renaming columns and constraints without recreating the table. It helps maintain and update the database as business requirements change.

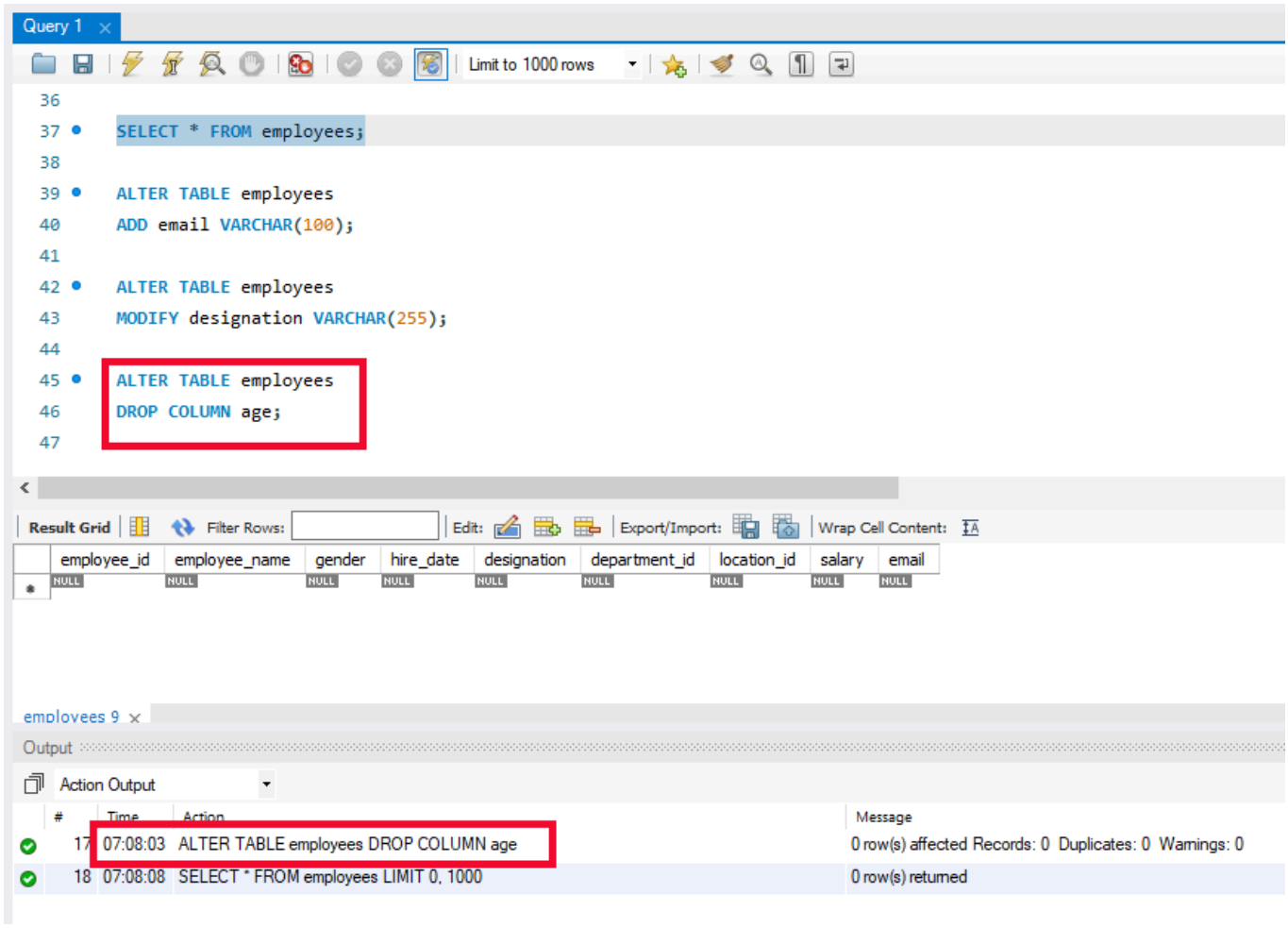
⦿ Add a new column named "email" to the Employees table to store employee email addresses.



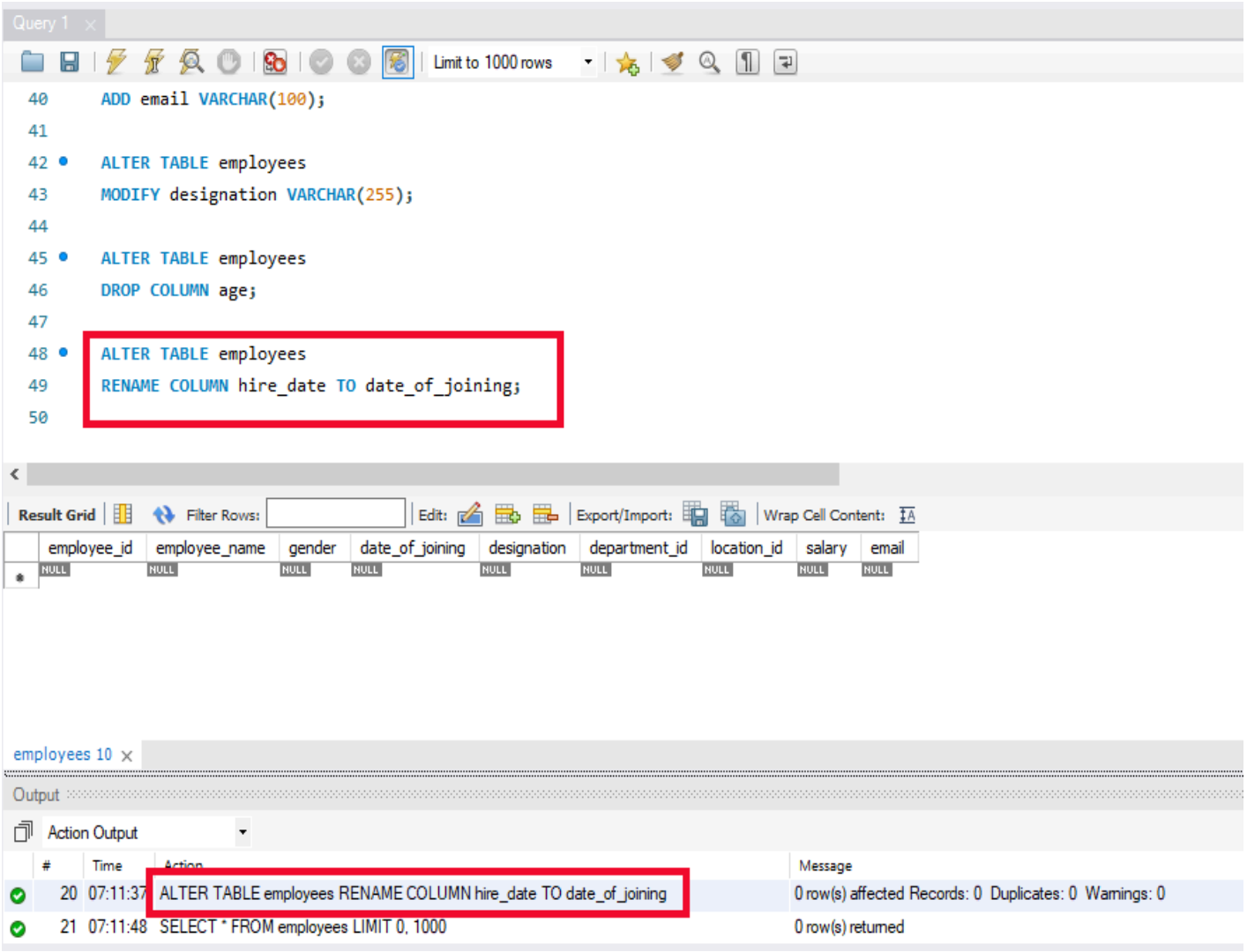
⦿ Modify the data type of the "designation" column in the Employees table to support a wider range of values.



⦿ Drop the “age” column from the Employees table.



⦿ Rename the “hire_date” column to “date_of_joining”.

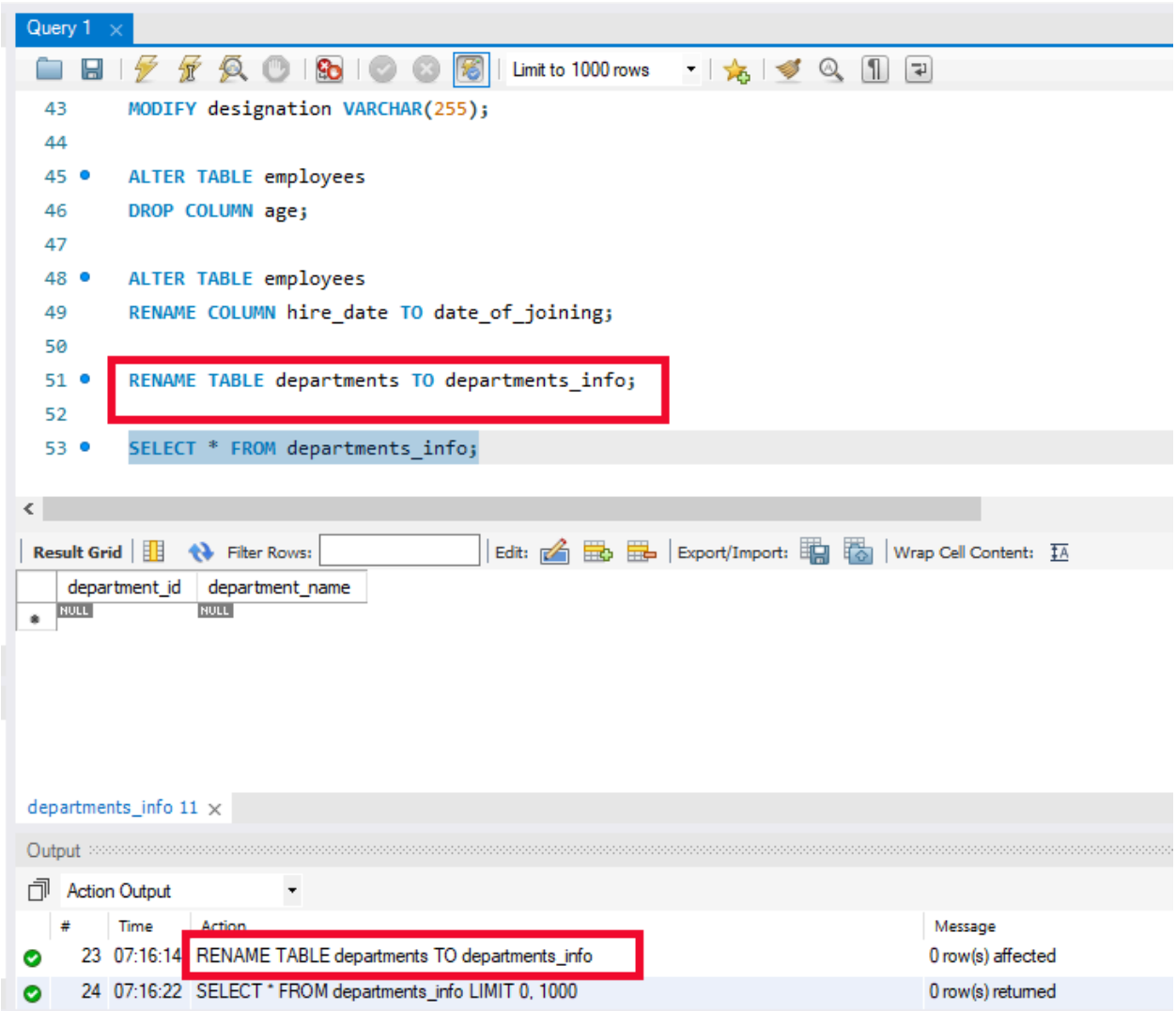


The **employees** table was altered to add an email field, modify the designation column size, remove the age column, and rename hire_date to date_of_joining.

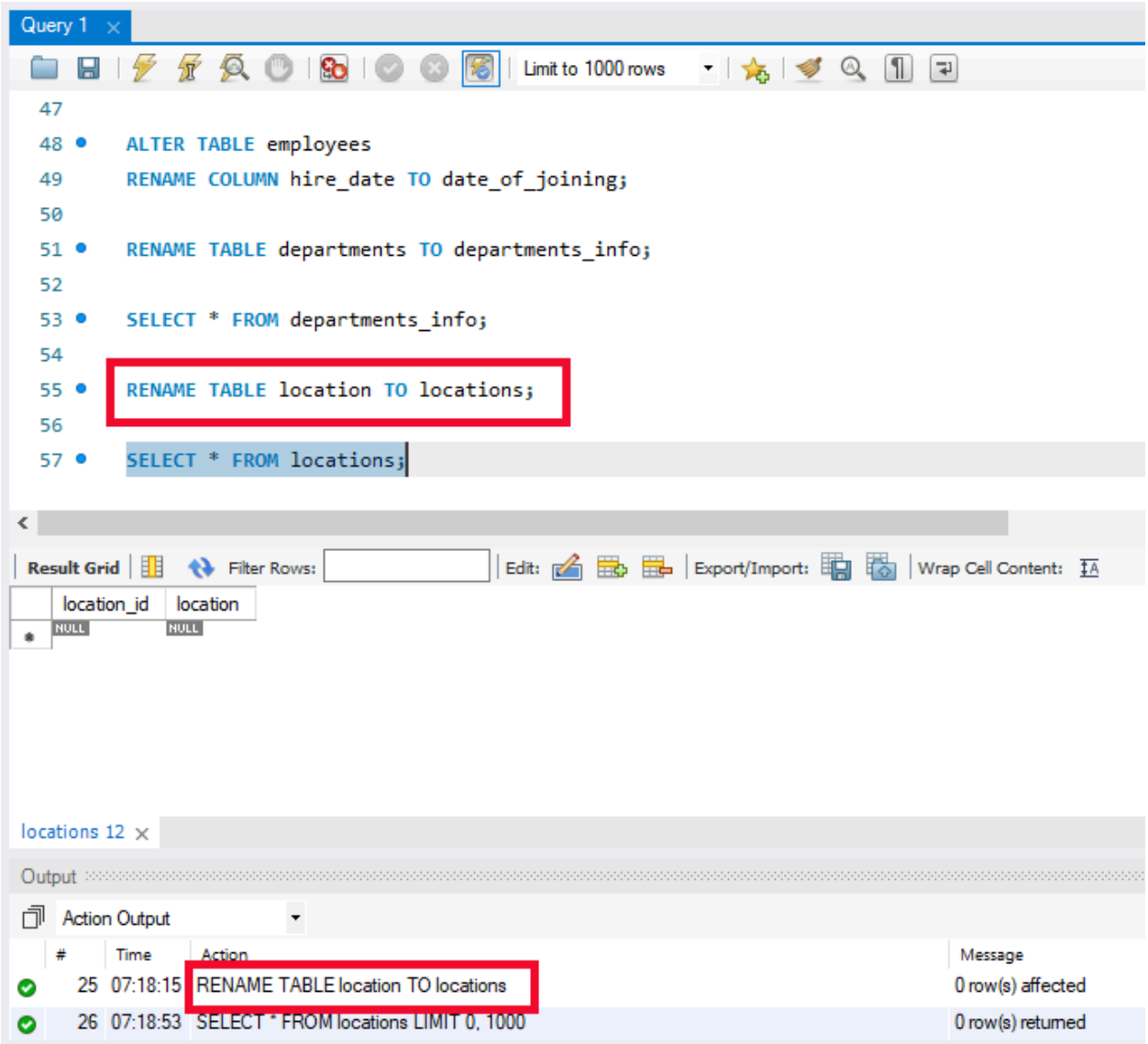
3. Table Renaming

The **RENAME TABLE** command is used to change the name of an existing table without affecting its data or structure. It helps improve database organization, follow naming standards, or reflect changes in business requirements.

⦿ Rename the "Departments" table to "Departments_Info".



⦿ Rename the "Location" table to "Locations".

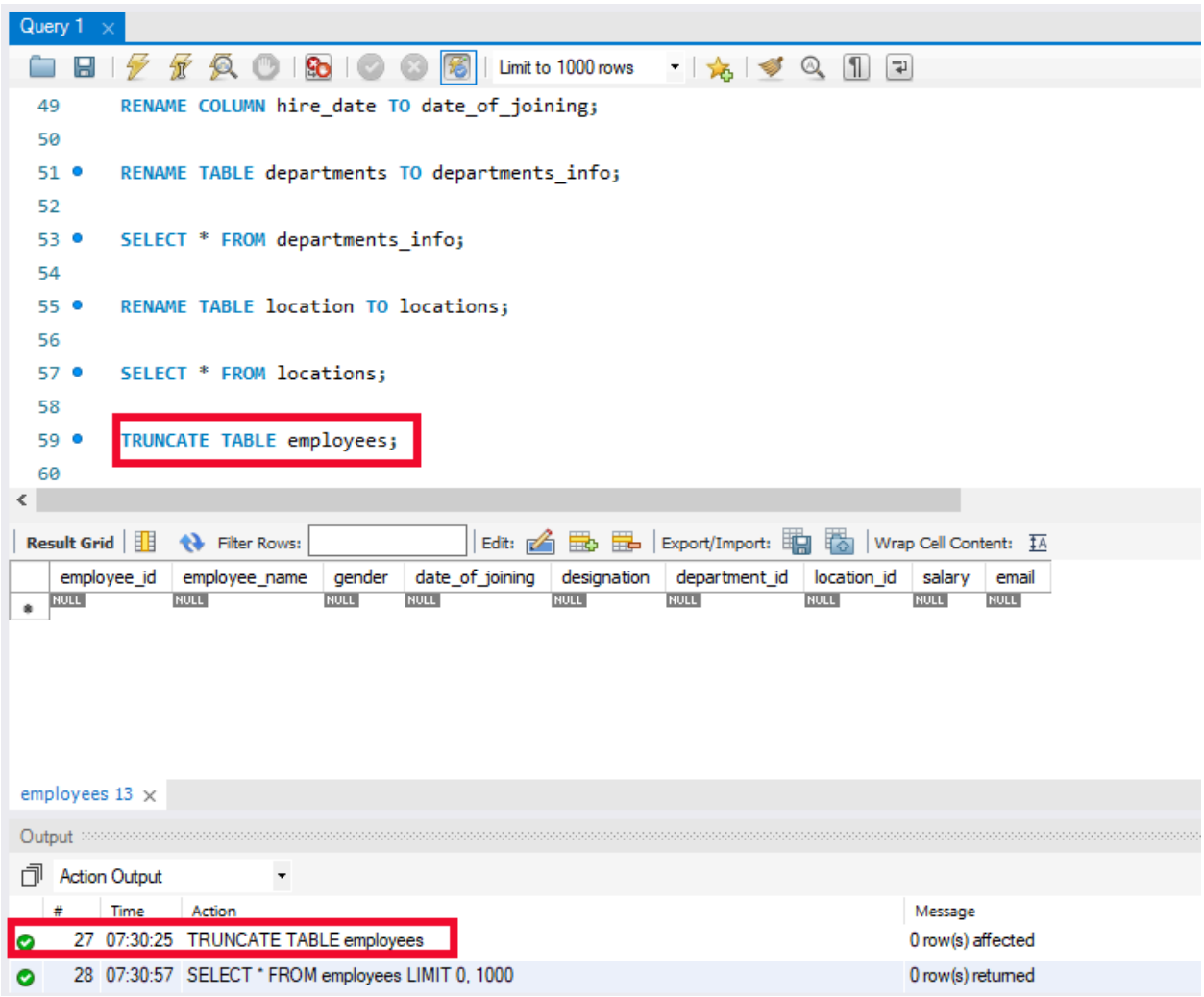


The tables departments and location were renamed to departments_info and locations respectively for better readability and standardization.

4. Table Truncation

Write an SQL statement to truncate the Employees table.

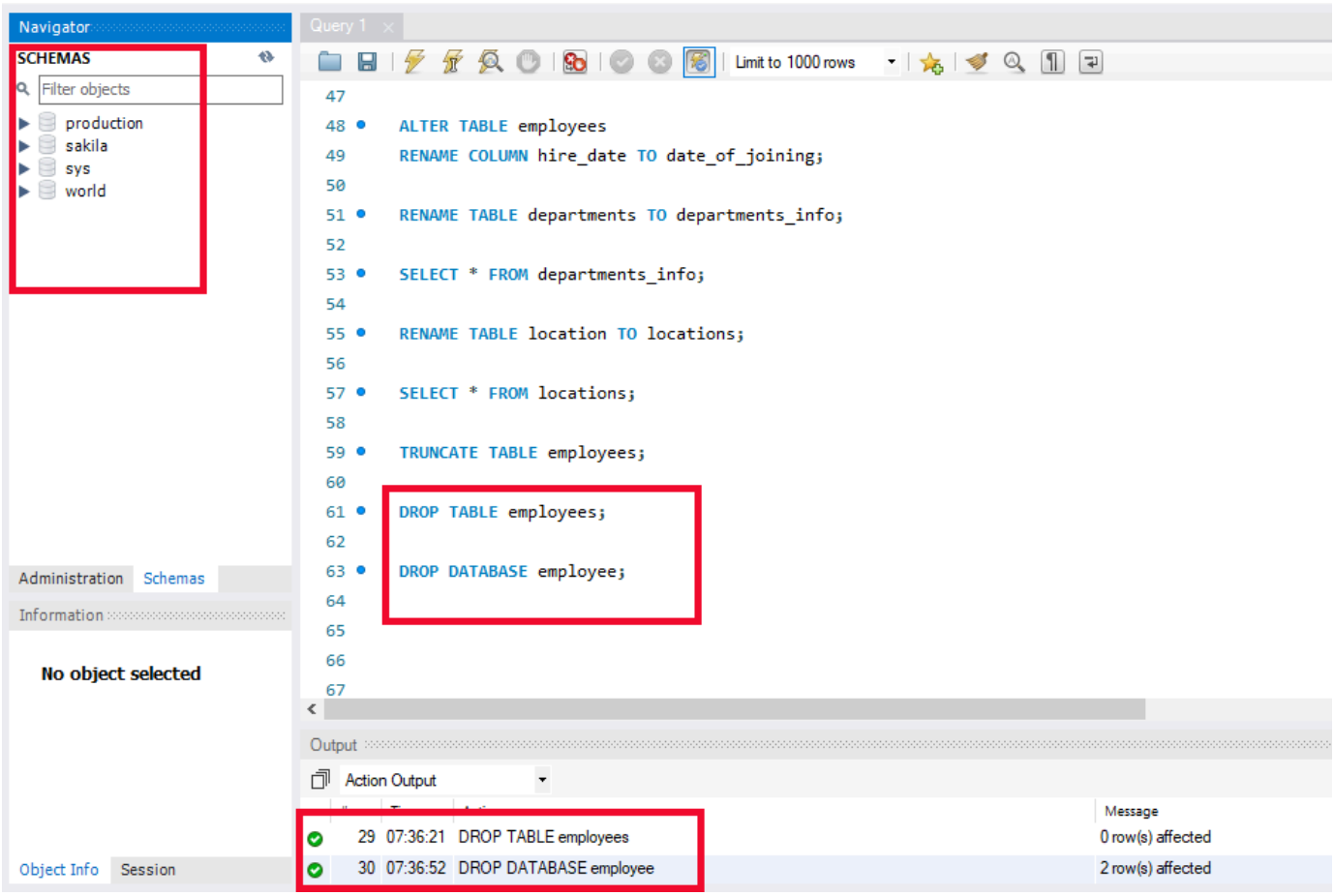
The **TRUNCATE TABLE** command is used to delete all rows from a table without deleting the table itself. The Employees table is truncated to clear all employee records while retaining the table definition.



5. Database & Table Dropping

Write the SQL statements to drop the Employees table and then the “employee” database.

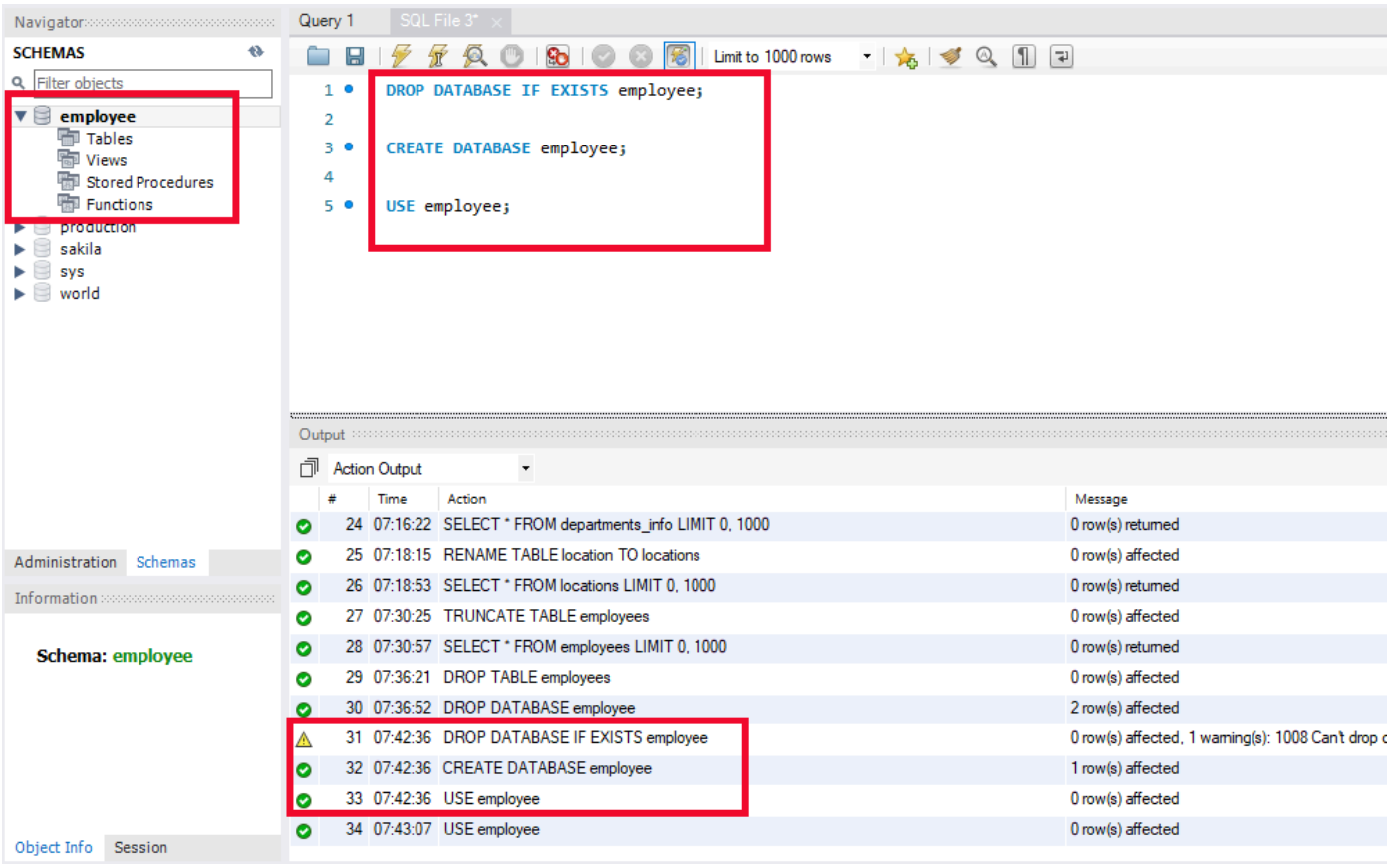
The **DROP** statement is used to permanently remove a database object such as a table, database from the system. The Employees table and the employee database are dropped using the DROP command.



Constraints :

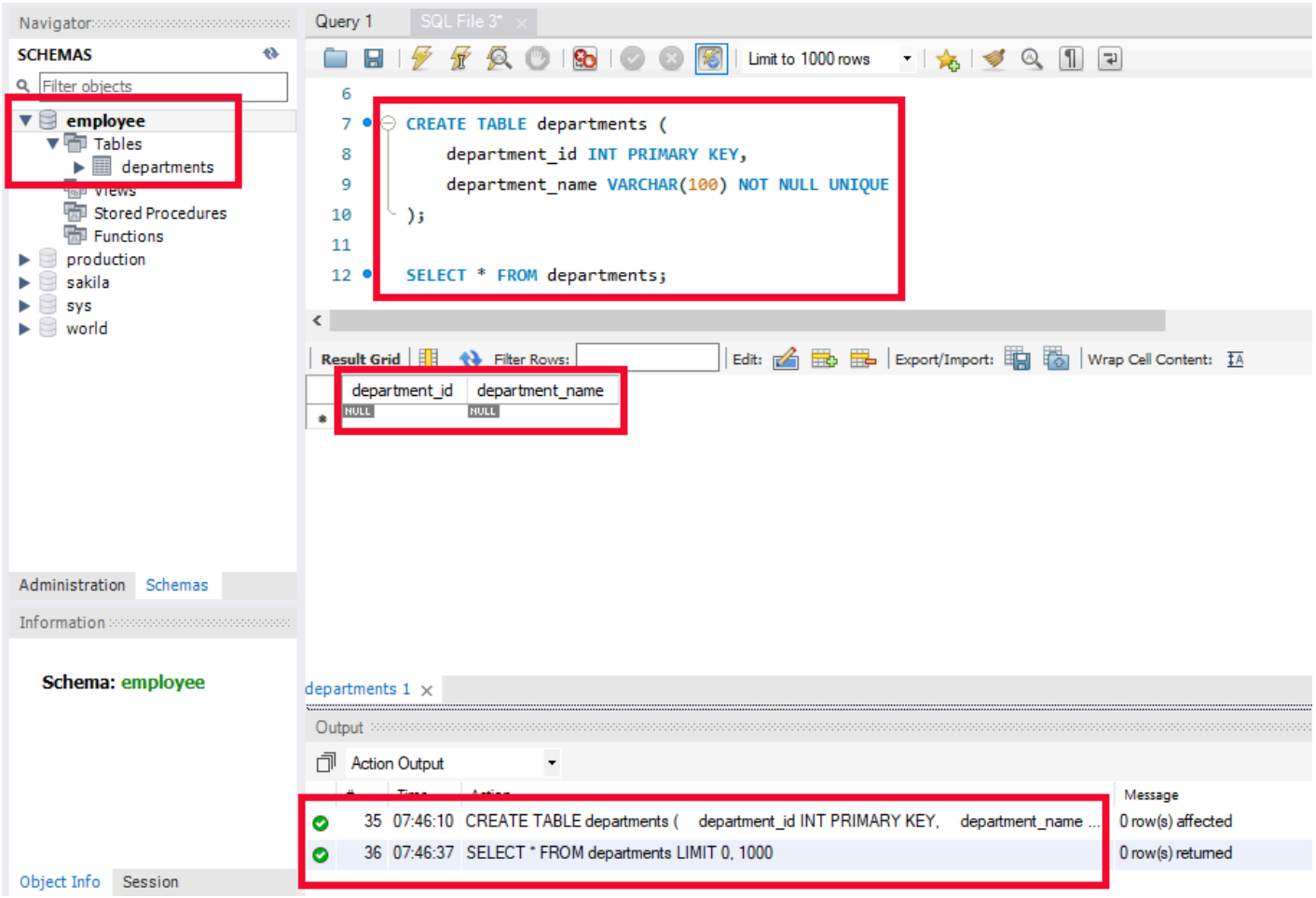
1. Database Recreation:

⦿ Drop the 'employee' database if it exists and recreate it using the provided schema, ensuring that all tables are created with the appropriate constraints as instructed.



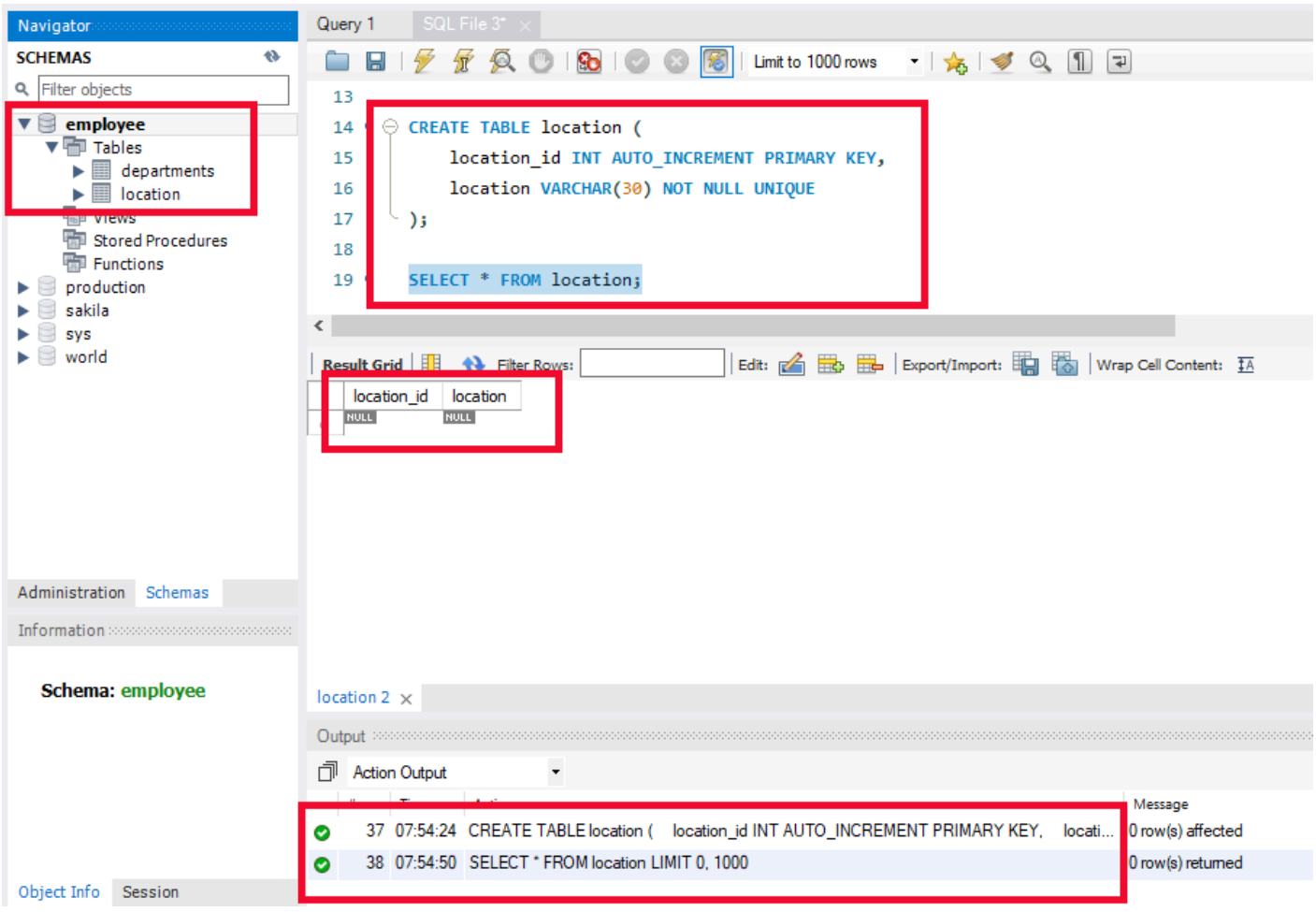
2. Departments Table:

⦿ Ensure that the "department_id" uniquely identifies each department.
⦿ Set up constraints on the "department_name" to avoid duplicate and null entries.



3. Location Table:

⦿ Establish a mechanism to automatically generate unique identifiers for each location, ensuring that they are incremented sequentially.
⦿ Implement constraints to prevent the insertion of null and duplicate locations.



4. Employees Table:

⦿ Guarantee that each employee has a distinct identifier.
⦿ Create a restriction to ensure that the employee's name is always provided.
⦿ Limit the acceptable values for the "gender" field to only 'M' or 'F'.
⦿ Enforce a condition to ensure that the employee's age is 18 or above.
⦿ Automatically assign the current date to the "hire_date" field if not specified.
⦿ Establish links between the "department_id" and "location_id" fields in the "employees" table and their respective tables.

